DATA ANALYTICS PORTFOLIO · 2026

Abdulrahman *Mohammed*

Data Analyst

Power BI · SQL · Excel · Python · Tableau

934/1000 — **Microsoft Certified: Power BI Data Analyst Associate**

PL-300 • Score: 934/1000 • Verify credential

About

Microsoft-certified Data Analyst combining an engineering background with professional experience in operations, education and Trust & Safety policy enforcement. Focused on business intelligence, operational performance and customer behaviour, with a portfolio spanning Dubai real estate, e-commerce delivery and retention. Analytical work integrates data modelling, SQL and statistical analysis, with an emphasis on data quality, metric selection and clear business interpretation.

How I work

01 **Frame** — Define the business question and scope.

02 **Validate** — Assess data quality and verify metric definitions.

03 **Analyze** — Investigate patterns and test assumptions.

04 **Explain** — Communicate findings and business implications.

01 P. 2

Dubai Real Estate Sales Performance & Trends

How did sales volume, value and price per m² change across 2023–2025?

526,056

sales transactions analysed, 2023–2025

Median price per m² chosen as the price KPI after a sensitivity test on 61 flagged records.

Power BI · Power Query · DAX · Star schema

VIEW PROJECT FILES

02 P. 4

Olist E-commerce & Delivery Analysis

How often were orders late, and how does lateness relate to reviews?

6.77%

of eligible delivered orders arrived late · 6,534 of 96,470

Late orders averaged 2.27/5 in reviews against 4.29/5 on time — an association, not proof of cause.

SQL (SQLite) · Power BI · DAX

VIEW PROJECT FILES

03 P. 6

Customer Journey & Retention Analysis

Where does the product journey lose visitors, and who returns?

2.12%

of eligible visitor-item pairs moved from view to cart within 7 days

Visitors with a first-day cart returned more often in days 1–7: 15.03% vs 5.63%.

Python · SQL (SQLite) · pandas · statsmodels

VIEW PROJECT FILES

Open to Data Analyst roles in *Dubai*

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01 Dubai real estate sales, *2023–2025*

Questions: How did sales activity, value and price per m² change over time — and did they move at the same rate? Which areas and property segments drove activity and value, and how did Existing and Off-Plan properties differ?

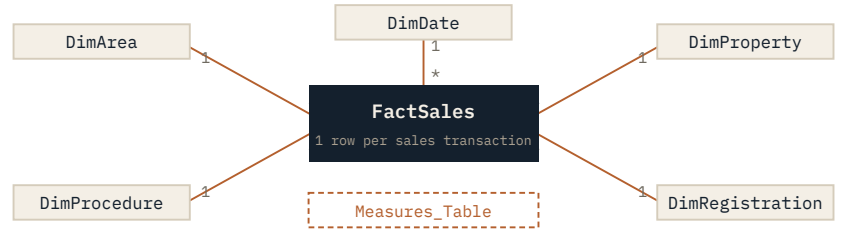
DATA	SCOPE	PERIOD	TOOLS
Dubai Real Estate Transaction dataset — a Kaggle mirror of **Dubai Land Department / Dubai Pulse** open transaction data, treated as a third-party extract	**526,056** sales transactions, one row per transaction ID · mortgage and gift records excluded · AED	**2023–2025** monthly and yearly	Power BI Desktop · Power Query · DAX · Performance Analyzer

PREPARE Appended the 2023, 2024 and 2025 extracts in Power Query, kept Sales only, set types, standardised text fields and **flagged suspicious records for review instead of deleting them**. Raw files stayed unchanged.

MODEL Star schema with five one-to-many, single-direction relationships and a dedicated measures table. Explicit DAX for volume, value, medians, shares and MoM/YoY growth.

VALIDATE Row count equals distinct transaction IDs; dimension keys match the fact table; totals reconcile by year (**131,805 · 179,425 · 214,826**); meter_sale_price matches value ÷ area within AED 1/m².

REPORT Four interactive pages — Market Overview, Market Trends, Areas & Property Analysis and an **Area Detail drillthrough** — checked with functional and visual QA and Performance Analyzer.



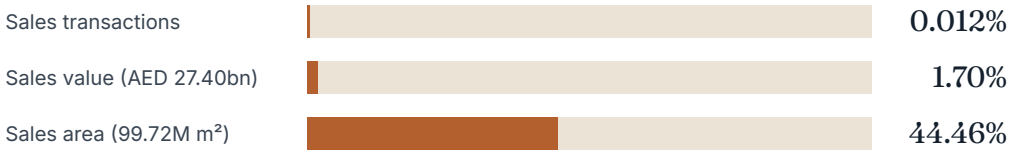
STAR SCHEMA, REDRAWN FROM THE POWER BI MODEL VIEW — **5 DIMENSIONS · 1-TO-MANY**

KEY ANALYTICAL DECISION

Why the price KPI is the median price per m²

A data-quality flag (**DQ_Flag = Review**) marked 61 records. They are a tiny share of transactions, but they hold almost half of all recorded sales area — enough to move any area-weighted price.

SHARE OF THE TOTAL HELD BY THE 61 FLAGGED RECORDS



Area-weighted price per m²



Fell by *about 44%*

MY CALCULATION FROM THE TWO VALUES

Median price per m²



Changed by *AED 0.67*

BOTH CHARTS SHARE ONE SCALE

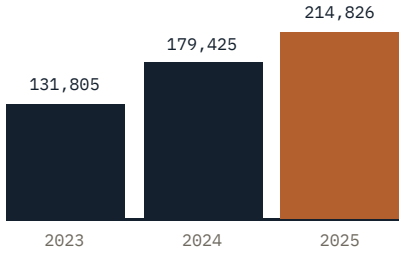
Decision: median price per m² is the primary price KPI. The flagged records stay visible in the core measures, and the area-weighted price is kept as a supporting diagnostic.

SALES TRANSACTIONS	TOTAL SALES VALUE	MEDIAN TRANSACTION VALUE	MEDIAN PRICE PER M²
526,056	AED 1,611bn	AED 1.55M	AED 16,745.31

FINDING 1 · GROWTH

Activity grew 63% — and value outpaced volume in 2025

Transactions rose **36.1%** in 2024 and **19.7%** in 2025. In 2024 volume grew faster than value; in 2025 the order reversed, with sales value up **31.2%**. Price movement alone should not be read as the cause, because property and area mix also shape total value.



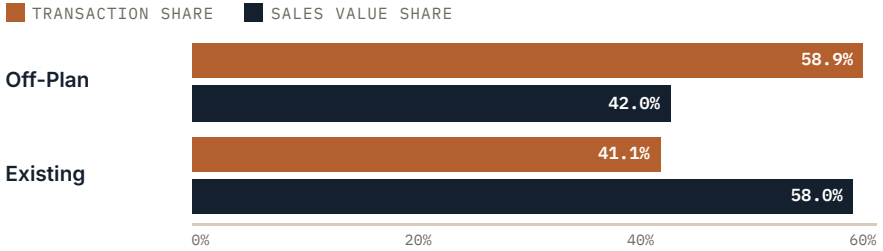
YOY	TRANSACTIONS	SALES VALUE	MEDIAN PRICE / M²
2024	36.1%	27.4%	11.7%
2025	19.7%	31.2%	8.4%

Median price per m² rose about 21.1% across the period. It describes the median recorded transaction, not a repeat-sales index.

FINDING 2 · MARKET MIX

Off-Plan led activity; Existing led value

Across 2023–2025, Off-Plan properties made up about **58.9%** of transactions but **42.0%** of sales value. Existing properties made up **41.1%** of transactions and **58.0%** of value. These shares describe market mix, not profitability.



FINDING 3 · AREAS

The busiest areas are not the highest-value areas

	TOP AREAS · TRANSACTIONS	TOP AREAS · SALES VALUE
1	**Al Barsha South Fourth**	**Marsa Dubai**
2	Business Bay	Business Bay
3	Wadi Al Safa 5	**Palm Jumeirah**
4	**Marsa Dubai**	Burj Khalifa
5	Madinat Al Mataar	Hadaeq Sheikh Mohammed Bin Rashid
6		**Al Barsha South Fourth**

Palm Jumeirah is outside the top ten by transactions. Rankings read from the Areas & Property Analysis page.



AREA DETAIL DRILLTHROUGH · AL BARSHA SOUTH FOURTH 2023–2025

The drillthrough shows the gap for a single area: Al Barsha South Fourth accounts for **9.6%** of transactions but **3.7%** of sales value, with a median transaction value of AED 887.50K against AED 1.55M for the whole market.

SUGGESTED NEXT STEPS

- Split median price per m² by property type and registration type to compare like with like.
- Review the 61 flagged records against the source before using area-based metrics.

LIMITS OF THE ANALYSIS

- Built on a non-official mirror extract, not reconciled to official DLD reports.
- Descriptive only: it shows what changed, not why, and is not a forecast or investment view.

02 Olist e-commerce: *sales, delivery and repeat buying*

Questions: How did orders and revenue develop? How often were delivered orders late, and how do late deliveries relate to review scores? Which categories and sellers stand out on delivery, and how many customers ordered again?

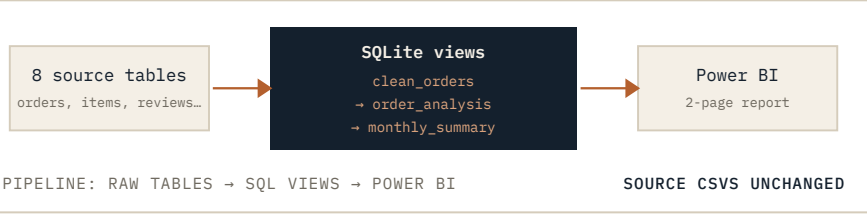
DATA
Brazilian E-Commerce Public Dataset by Olist (Kaggle) — real marketplace orders from Brazil · CC BY-NC-SA 4.0

SCOPE
99,441 orders · 96,478 delivered · 96,470 with both delivery dates

PERIOD
2016–2018
trend charts Jan 2017 – Aug 2018

TOOLS
SQLite · DB Browser for SQLite · Power Query · DAX · Power BI Desktop

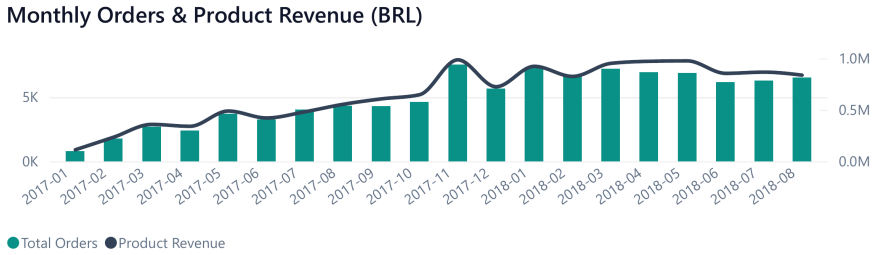
PREPARE	Imported eight Olist tables into SQLite with IDs kept as text. Built three views: **clean_orders** parses timestamps; **order_analysis** keeps one row per order, aggregating items, payments and reviews by order before joining so nothing is double-counted; **monthly_summary** feeds the trend.
ANALYSE	SQL queries for monthly orders and revenue, the overall late rate, reviews for late versus on-time orders, top categories and sellers, and repeat customers by customer_unique_id.
MODEL	Power BI model with a date table, orders, order items and products in one-to-many, single-direction relationships. **TREATAS-based measures** pass distinct order IDs back to the order table, so category and seller late rates work without bidirectional filters.
REPORT	Two pages — **Commercial Overview** and **Delivery & Customer Experience** — with a purchase-date slicer. Monthly charts show Jan 2017 – Aug 2018; sparse edge months stay in the full-period KPIs.



KEY ANALYTICAL DECISIONS

Definitions that decide the numbers

Late delivery	Actual delivery **calendar date** after the estimated date; later on the same day counts as on time.
Eligible delivery	Delivered with both dates: **96,470** of 96,478. Eight orders without a required date are excluded from rates.
Product revenue	Item prices on delivered orders, **excluding freight** — not Olist's commission or net revenue.
Review score	Averaged per order first, then across reviewed orders.
Seller / category	Distinct orders per group. A multi-seller order appears in each group, so **groups are not additive**.



ORDERS AND REVENUE GREW SUBSTANTIALLY DURING 2017

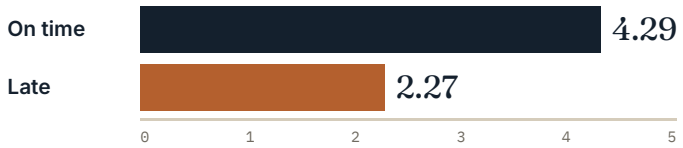
COMMERCIAL OVERVIEW PAGE

TOTAL ORDERS	PRODUCT REVENUE	AVG DELIVERED ORDER	LATE DELIVERY RATE	REPEAT CUSTOMERS
99,441	BRL 13.22M	BRL 137.04	6.77%	3.00%

FINDING 1 · REVIEWS

Late orders scored about 2 points lower

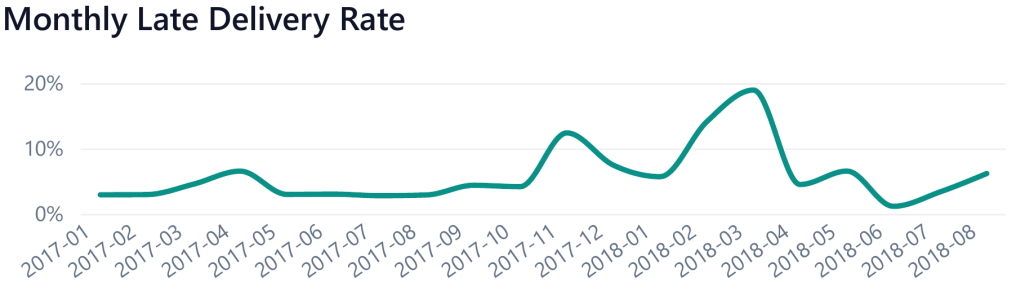
6,534 of 96,470 eligible deliveries arrived late. Late orders averaged **2.27/5** in reviews against **4.29/5** for on-time orders.



An association, not evidence that delay alone caused the difference.

FINDING 2 · TIMING

The late rate peaked around March 2018



Monthly rates use the purchase month. Identifying the cause of the peak needs further investigation.

FINDING 3 · SELLERS AND CATEGORIES

Delivery performance varies across high-volume sellers

Among the five highest-volume sellers, late rates ranged from **5.27%** to **9.71%**; among the top five categories, from 6.39% to 7.51%.

TOP 5 SELLERS · ELIGIBLE ORDERS

SELLER ID	ORDERS	LATE	RATE
6560211a…	1,819	96	5.28%
4a3ca931…	**1,772**	**172**	**9.71%**
cc419e06…	1,651	87	5.27%
1f50f920…	**1,399**	**124**	**8.86%**
da8622b1…	1,311	87	6.64%

TOP 5 CATEGORIES · ELIGIBLE ORDERS

CATEGORY	ORDERS	LATE	RATE
Bed Bath Table	9,272	689	7.43%
Health Beauty	8,647	649	7.51%
Sports Leisure	7,529	495	6.57%
Computers Accessories	6,529	417	6.39%
Furniture Decor	6,307	449	7.12%

From the report's top-10 tables. A late order does not show which seller or carrier caused the delay.

FINDING 4 · REPEAT BUYING

3.00%

2,801 of 93,358 customers with a delivered order placed another within the period.

Within the dataset window only — not a lifetime retention measure.

SUGGESTED ACTIONS

1. Investigate the March 2018 spike **by seller and route** before changing delivery promises.
2. Prioritise sellers with both **high volume and elevated late rates**, reading counts alongside percentages.
3. Test a post-delivery follow-up for first-time buyers and measure repeat buying over a fixed window.

– Historical data (2016–2018); not current marketplace performance.

03 Customer journey: *where visitors drop off*

Question: Where does the observed product journey lose the most visitors, and which early behaviour is associated with returning?

DATA

Retailrocket recommender system dataset (Kaggle) — anonymized events from a real e-commerce site · CC BY-NC-SA 4.0

SCOPE

2,756,101 view, add-to-cart and purchase events · 2,755,641 after cleaning

PERIOD

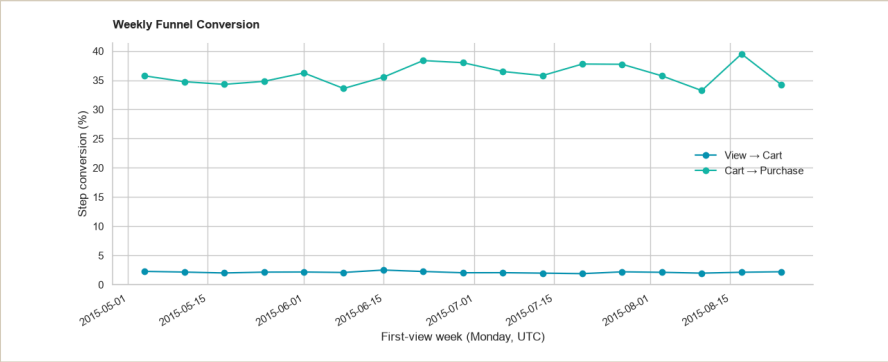
3 May – 18 Sep 2015 UTC

TOOLS

Python · pandas · SQLite · statsmodels · Matplotlib · Seaborn · Jupyter

CLEAN	Confirmed no missing required fields and no purchase without a transaction ID. Removed **460 exact duplicate rows**, assuming they were duplicate logs; the source CSV is unchanged.
FUNNEL · SQL	Built an ordered view → cart → purchase funnel in SQLite, then grouped it by first-view week to check consistency across **17 complete weeks**.
RETENTION · PYTHON	Weekly cohorts by first observed week; each later week counts distinct active visitors over the original cohort size. **Periodic, not cumulative**; unfinished weeks stay blank rather than zero.
TEST · STATSMODELS	Compared days 1–7 return for visitors with and without a first-day cart: two-sided pooled **two-proportion z-test** and a 95% Newcombe interval. Observational, not an A/B test.

Two different units: the funnel counts **visitor-item pairs**; retention and the return test count **visitors**.

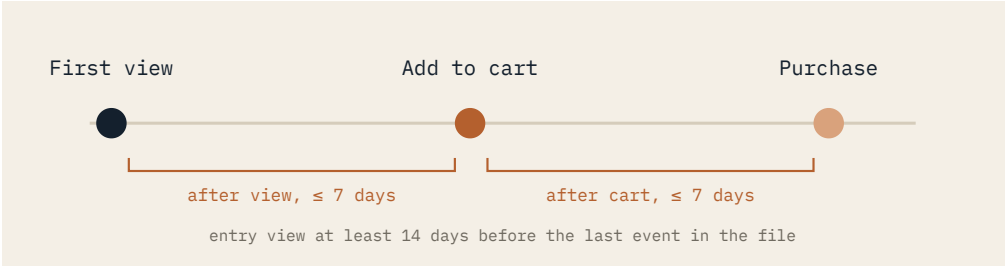


WEEKLY STEP CONVERSION · 17 COMPLETE WEEKS NOTEBOOK OUTPUT

KEY ANALYTICAL DECISION

An ordered, time-bound funnel — not raw event counts

Each visitor-item pair is counted once, anchored to its first observed view, and each step must follow the previous one within a fixed window.



Why first view	Uses one first observed view per pair **rather than all raw view events**, so repeat views don't inflate the base.
Why strict order	Cart must come after the view and purchase after the cart; equal timestamps don't prove order.
Why 14 days	Every entry pair has a full follow-up window, which reduces end-of-file censoring.
What's excluded	Purchases without an observed view → cart sequence. Purchased pairs are not unique orders.

FINDING 1 · FUNNEL

97.88% of eligible product views did not reach cart



VIEW → CART

2.12%

CART → PURCHASE

36.05%

VIEW → CART WAS THE WEAKER STEP IN 17 OF 17 COMPLETE WEEKS



Bars and labels are shares of all eligible viewed pairs; weeks run 4 May – 24 Aug 2015.

FINDING 2 · RETENTION

Return activity falls quickly after the first week

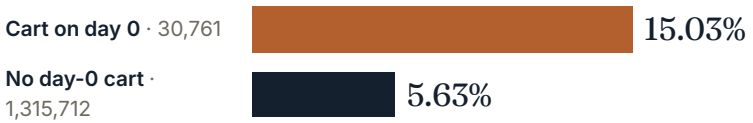
For visitors first seen in the week of 4 May, about **4.8%** returned in week 1, **2.8%** in week 2 and **1.6%** in week 4.

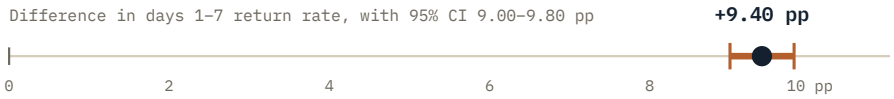
Cohort week	W1	W2	W3	W4	W5	W6	W7	W8
4 May	4.8	2.8	2.1	1.6	1.3	1.3	1.1	1.0
11 May	4.3	2.1	1.7	1.2	1.1	1.0	0.8	0.9
18 May	3.5	1.9	1.3	1.1	1.0	0.8	0.8	0.8
25 May	3.4	1.6	1.2	1.0	0.9	0.9	0.8	0.8
1 Jun	3.4	1.8	1.3	1.0	0.9	0.9	0.8	0.7
8 Jun	3.4	1.7	1.2	1.1	0.8	0.8	0.6	0.6

% of each cohort active in that week, rounded to 0.1 — read from the notebook's cohort chart. The partial 27 April cohort is not shown.

FINDING 3 · EARLY BEHAVIOUR

First-day cart visitors returned more often





Two-sided pooled z-test, $p < 0.001$. Observational: early cart visitors may already have stronger intent, so this is not the effect of a cart prompt.

RECOMMENDATIONS

1. **Investigate view-to-cart first** — product pages, tracking, availability and traffic source — then test one change against a randomized holdout.
2. **Test a first-week return reminder** where consent exists, with days 1–7 return as the primary outcome and purchases as a guardrail.
3. **Track first-day cart segments separately**, using the +9.40 pp gap as a segmentation signal, not a promised impact.

- Anonymous browser IDs may not map one-to-one to people; no bot or abnormal-user filter was applied.
- 2015 behaviour is not a benchmark for current Dubai or Gulf conversion.

Let's talk — Data Analyst roles in *Dubai*

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